

01 MEASURE

Stems, basal area, heights and sound organ volume

Paired material density and carbon fraction

Independent plots; fixed epoch and class

02 RECONSTRUCT

Carbon / ha = $\beta \times G \times \Lambda$ + remainder

Λ = weighted height $\times$ weighted architecture

Average plot products; retain covariance

03 COMPARE

Independent measured minus mapped carbon

R < 0 can support an overestimated map

R > 0 can support an underestimated map

CLASS 5 · 531.45 MILLION HECTARES · ABOUT 5.31 MILLION km²

Comparable in scale to the historical Amazonian forest extent; not the same domain or a current area estimate.

Density basis	MgC/ha	Mg dry/ha	m³/ha
Class 5: lower class edge	478	≈ 1,020	≈ 1,700
Class 5: printed mean	531.05	≈ 1,130	≈ 1,880
Class 5: stock / area	536.33	≈ 1,140	≈ 1,900
Classes 4 and 5	472.19	≈ 1,000	≈ 1,670
Illustrative f = 0.47; ρ = 0.60 Mg/m³; separate remainder = 0.			

Fixed biomass denominator	Printed	Ratio
African mean: 395.7 Mg/ha	2.86×	2.88×
Amazonian value: 350 Mg/ha	3.23×	3.26×
Amazonian value: 200 Mg/ha	5.65×	5.71×
Factors compare different populations and estimators. Each column changes only the class-5 convention.		

Model-class edge 478: applies to assigned cell estimates if the class rules are implemented; not every field plot or subpixel patch.

At weighted height 30 m: G = 30 / 40 m²/ha requires weighted $\psi \approx 2.1$ / 1.6 at the ratio target. These are scenarios, not observed stands.

FIVE EXPLANATIONS AND THE OBSERVATIONS THAT DISCRIMINATE THEM

1 Solid volume / architecture	Measure sound stems, branches, taper and cavities independently. A large ψ inferred from the carbon target does not test that target.
2 Material properties	Measure paired density and carbon fractions across organs and strata. Lower β raises required wood volume; β is not an adjustable fit coefficient.
3 Remainder compartments	Foliage $\approx$ 1.05% of dry mass in a harvested-tree subset; live epiphytes $\approx$ 2% in one stand. A 65% carbon remainder is unsupported by these examples. Woody lianas stay in woody accounting.
4 Spatial population selection	Match the class map to independent, representative field strata and sampling weights. Different forest populations can differ physically; a selected exceptional hectare is insufficient.
5 Mapped target overestimated	Test map calibration, out-of-domain prediction and Baccini/Simard predictor discrimination. Held-out plots and negative residuals beyond a prespecified margin discriminate upward map bias.

DECISION: compare a joint-uncertainty interval for R with a prespecified equivalence margin in both directions.

Current empirical application: Córdoba aggregates are reproduced; independent volume and matched map values are missing, so R is unidentified.

Foliage and epiphyte examples are population-specific. A 10% remainder screen is an explicit scenario, not a universal ecological ceiling.